

Bio-inspired Algorithms for Loan Default Detection

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Abstract—This study assesses five hyperparameter optimization algorithms: Genetic Algorithm, Random Search, Bayesian Optimization, Particle Swarm Optimization, and Grid Search to improve loan default prediction models. Using the Default of Credit Card Clients dataset, this study optimized four machine learning classifiers: Decision Tree, Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), and Support Vector Machine (SVM). The results demonstrate that bio-inspired, population-based algorithms: Particle Swarm Optimization (PSO) and Genetic Algorithm (GA) outperformed traditional search methods in identifying optimal hyperparameter combinations. Gradient boosting intensified with XGBoost, optimized using a Genetic Algorithm, showed the strongest predictive ability for identifying loan defaults. Comparing all results, this paper concludes that evolutionary algorithms outperform traditional optimization approaches when fine-tuning financial models.

Index Terms—Evolutionary Algorithms, Machine learning, XGBoost, LightGBM, Loan Default Prediction

I. INTRODUCTION

Lending is the lifeblood of global economic expansion, providing borrowers with money to invest, expand and spend. Nevertheless, loan defaults are a worry as they not only affect the liquidity of financial firms but create enough shocks and mayhem at such large quantities that can create systemic risk in the whole market. Hence, correct credit risk predictions are crucial in maintaining profitable lending and ensuring that financial system is stable and non-performing assets (NPAs) are minimized.

Traditional credit scoring models (such as logistic regression or expert based) usually do not account for non-linear relationships and detailed borrower behaviours. Models like XGBoost are powerful, but their performance is highly dependent on hyperparameters settings (e.g. learning rate, tree depth, regularization strength). These parameters dictate how a model learns, balance between the bias and variance with trade-offs, and generalize to new data. However: The space

to search is a high-dimensional, complex, and uncertain one. Manual tuning is therefore slow, ineffective and frequently not repeatable. Small misconfigurations can result in underfitting, overfitting and instability.

Therefore, the demand for automatic hyperparameter optimization using various algorithms such as Genetic Algorithms, Bayesian Optimization, Grid Search/Random is important to systematically find the best parameter setting and get high model performance.

Optimization paradigms refer to various strategies or methods used to find the best settings for a model. Each paradigm balances accuracy, speed, and computational cost differently [1]. Grid Search thoroughly tests all feasible combinations of parameters within a defined search space [2]. Random Search, on the other hand, selects a few combinations at random from the search space [3]. Bayesian Optimization is a smarter approach to optimization. It utilizes the concept of a surrogate model to predict the quality of combinations of hyperparameters [4]. Genetic Algorithms and Particle Swarm Optimization are popular approaches that mimic nature's biology and social interactions. GA imitates natural evolution, where solutions evolve over generations through operations like selection, crossover, and mutation [5]. PSO models the social movement of birds or fish [6].

This paper is structured as follows. Related Work Section II gives an overview of existing literature on loan-default prediction and hyperparameter optimization. The methodology, which also includes data preprocessing, feature engineering and a mathematical formulation of the optimization algorithms employed is described in Section III. The experimental results are discussed and the performance of the models is compared with various optimization methods in Section IV. Section V presents possible future research directions, while Section VI summarizes the study by restating the findings.

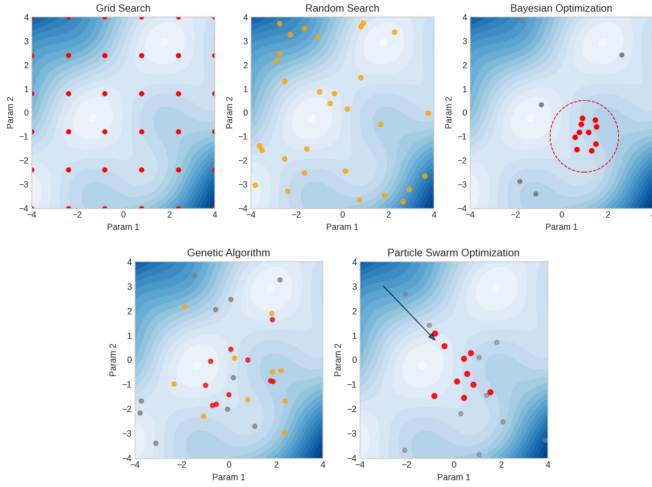


Fig. 1. Hyperparameter Search in Different Optimization Methods

Figure 1 shows how different methods explore the hyperparameter space: Grid Search evaluates a fixed grid of points, Random Search samples points randomly, Bayesian Optimization focuses on promising regions, Genetic Algorithm evolves solutions toward better areas, and Particle Swarm Optimization moves candidate solutions collectively toward optimal regions.

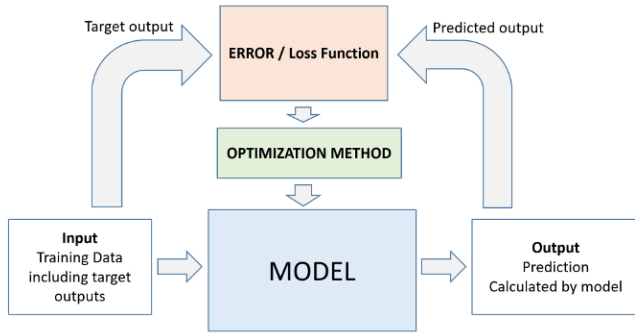


Fig. 2. Flowchart of Optimization Algorithms [7]

Figure 2 shows how a model learns by taking training data, producing predictions, comparing them with true targets through a loss function, and using an optimization method to adjust its parameters. This loop repeats until the model improves its accuracy.

II. LITERATURE REVIEW

Loan default prediction is one of the critical tasks in the financial industry that directly influences lending decisions and risk management strategies [8]. Traditional statistical methods such as logistic regression are inflexible, which limits the ability of these methods to account for nonlinear relationships in financial data [9]. Advanced machine learning techniques, such as tree-based and deep-learning models, have been demonstrated to be able to manage the aforementioned

non-linear patterns more effectively and provide a higher level of prediction accuracy [10]. Bio-inspired optimization algorithms like GA and PSO that mimic natural processes like evolution and swarm intelligence, respectively, are often observed improving the prediction accuracy and convergence efficiency [11]. Consequently, researchers have turned to bio-inspired algorithms to get the desired optimal hyperparameters.

A. Review on Existing Studies for Loan Default Detection

TABLE I
REVIEW ON EXISTING STUDIES FOR LOAN DEFAULT DETECTION

Paper Title	Author Names	Work Done	Technique Used	Results
Loan Default Risk Prediction Using Knowledge Graph [12]	Md. Nurul Alam et al.	Predicting loan default risk using Knowledge Graph Embeddings (KGE).	Knowledge Graph, XGBoost.	Accuracy: 96.79% using XGBoost + KGE.
A Novel Predictive Model for Housing Loan Default using Feature Generation and Explainable AI [13]	Mohamed Mahyoub et al.	Predicting housing loan default using feature generation and explainable AI.	Feature generation, SHAP, LIME.	Accuracy: 93% using XGBoost.
Loan Default Prediction Using Machine Learning Techniques and Deep Learning ANN Model [14]	A. Lakshmanarao et al.	Comparative study of ML and DL models.	Various ML techniques, ANN.	ANN outperforms all proposed ML techniques.
Research on Loan Default Risk Prediction Methods Based on Graph Neural Networks [15]	Heguang Chen et al.	Loan default prediction using graph neural networks.	ML, CNN, graph-based methods.	GraphSL outperforms standard graph methods.
Loan Default Prediction with Machine Learning Techniques [16]	Lili Lai et al.	ML models for corporate and consumer loan defaults.	XGBoost, AdaBoost.	Improved accuracy and F1-score over logistic regression.
Effective Machine Learning-Based Predictive Analytics for Loan Default Prediction in Banking Sector [17]	Dr. Yogesh Kumar Jain et al.	Comparison of machine learning models.	Random Forest.	Random Forest delivered superior results.

B. Research Gaps

- Conventional hyperparameter optimization relies on standard search methods; in this paper, evolutionary optimization is employed to enhance exploration and solution quality.
- Traditionally, hyperparameter optimization (HPO), and feature selection have been treated as two separate and sequential processes. In contrast to this method of treating HPO independently of feature selection, this study develops a unified framework that jointly optimizes HPO and feature selection to maximise performance and efficiency of models.
- To accomplish this, this paper reformulates HPO as a multi-objective optimization problem to provide a balance between competing goals (e.g., performance versus complexity). The outcome of this framework provides a Pareto front of solutions rather than the classical single-optimal approaches, thus providing a mechanism for stakeholders to evaluate the trade off between competing objectives for realistic deployment in industry.

III. METHODOLOGY

This approach revolves around the comparison of five hyperparameter optimization approaches (PSO, Bayesian, Grid Search, and Random Search) to predict loan default. The “Default of Credit Card Clients” dataset was taken and cleaned via a series of preprocessing steps, including data cleaning, feature engineering, and Synthetic Minority Over-sampling Technique (SMOTE) balancing. These optimization methods were used to tune four different machine-learning models, DTC, SVM, LightGBM, and XGBoost. A final comparison of the model-optimizer pairs was performed to get the best performing combination.

A. Data Preprocessing

Dataset Description: The dataset utilized in this study was obtained from the University of California, Irvine (UCI) [18], a well-established open-source platform for standardized machine learning datasets. Prior research has emphasized the importance of detailed borrower behaviour variables [19]. The dataset offers a complete profile of the borrower (demographics, financial figures [credit limits], and repayment history). It follows payment reliability (PAY_0–6), in addition to bill statements (BILL_AMT1–6) and the actual payments (PAY_AMT1–6). The target variable is the loan status.

The dataset underwent preprocessing steps wherein missing values were handled, duplicates were removed, feature normalisation and encoding of categorical variables was taken care off.

Data Cleaning: The dataset was cleaned in order to ensure its quality and was prepared for modelling. Repeated records were excluded to prevent bias in model training. Numerical variables with missing values were imputed using the median. Accurate default prediction has been documented as requiring high-quality loan data [20].

Feature Engineering: Binary encoding was applied to two-category variables, while target encoding was used for multi-category variables. Numerical features were standardized through scaling to ensure equal contribution during training and to prevent bias from large-valued features. Feature selection has been shown to enhance loan default detection [21].

Handling Data Imbalance: The dataset’s 80–20 class imbalance was addressed through random undersampling of non-default instances and SMOTE oversampling of default instances, enhancing model fairness and predictive performance.

B. Model Building

Four different supervised machine learning algorithms were trained for the prediction of loan defaults: DTC, SVM, LightGBM, and XGBoost. DTC was used as it allows for the combination of categorical and numerical data. The SVM model was also trained since it works well in high-dimensional space and can construct optimal separation boundaries with the use of kernel functions. Both LightGBM and XGBoost were used since gradient boosting ensemble methods can capture complex nonlinear relationships as well as improve models through iterative tree learning when ensemble trees are added to the model. Several recent studies have demonstrated the effectiveness of machine learning techniques for loan default prediction, particularly when combined with systematic hyperparameter tuning [8], [22]. The combination of these models provides a good balance of simplicity, interpretability, and high predictive power.

Train-Test Split: The preprocessed dataset was partitioned into training (70%) and test (30%) sets. Stratified splitting was used to preserve the class distribution between the sets, and a fixed random seed was set to ensure reproducibility.

Feature Selection:

- **Filter methods:** Univariate statistical tests such as chi-square and mutual information were applied along with correlation filtering using Pearson and Spearman thresholding to remove irrelevant or highly collinear features.
- **Wrapper methods:** Recursive Feature Elimination (RFE) with cross-validation and evolutionary algorithms (e.g., Genetic Algorithm) were applied to identify the optimal subset of features that improve model performance.

Model Implementation: Four supervised machine learning models—DTC, SVM, LightGBM, and XGBoost—were implemented and trained using the selected feature subsets.

Optimization Techniques: Five commonly used hyperparameter and feature search strategies were compared in this study. Genetic Algorithm is an evolutionary approach that enhances candidate solutions through selection, mutation, and recombination [23]. Particle Swarm Optimization is based on the collective motion of candidate solutions, where positions are updated according to both individual and group performance [24]. Bayesian Optimization leverages probabilistic surrogate models such as Gaussian Processes or Tree-Structured Parzen Estimators to balance exploration and exploitation during the search process [4]. Grid Search evaluates all parameter combinations within a predefined search space, while Random

Search samples parameter settings randomly to explore a wider range of configurations [25].

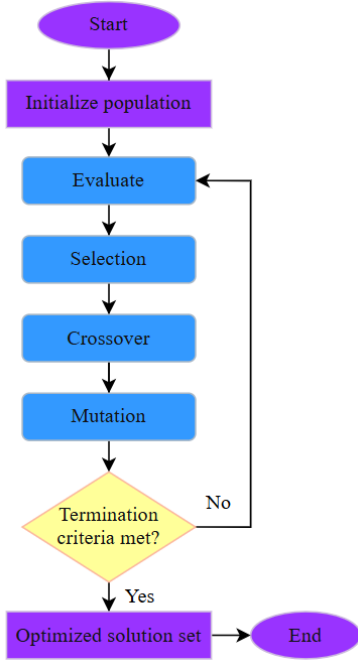


Fig. 3. Flowchart of Genetic Algorithm

Figure 4 indicates flowchart of Genetic Algorithm as an evolutionary search which initiates from a population of potential solutions and evolves them through generations. It applies selection to keep the best, crossover to merge useful features from candidates, and mutation for adding new variations. By these cycles of operations, the algorithm can drive the population smoothly toward better solutions and are applied for difficult search and optimization problems.

C. System Architecture

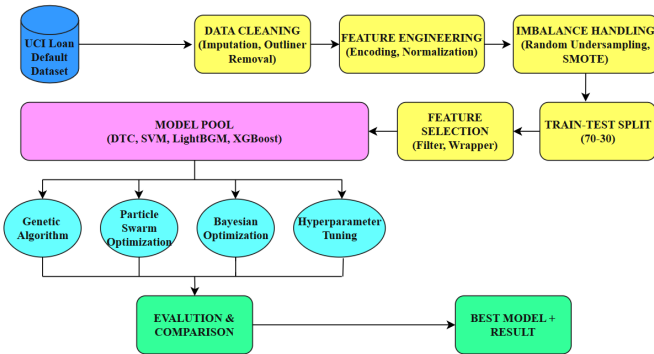


Fig. 4. System architecture

The system architecture (Fig. 5) starts from the UCI Loan Default Dataset and it goes through data preprocessing (imputation, outlier removal) and feature extraction (encoding, normalization). The dataset has an imbalanced class distribution,

which is mitigated by Random Undersampling and SMOTE before splitting the dataset into 70:30 train-test splits. The most important features are extracted through the Filter and Wrapper methods, which serve as inputs for a model pool consisting of DTC, SVM, XGBoost, and LightGBM. These models' individual parameters are optimized using Genetic Algorithm, PSO, Bayesian optimization and standard tuning. Eventually, the models are evaluated and compared to choose a classifier with better performance.

D. Mathematical Formulation of Optimization Algorithms

Mathematical formulation of optimization algorithms provides a systematic framework for finding efficient solutions to complex problems by elaborating objective functions, constraints, and decision variables mathematically. Here are the mathematical formulation of the optimization techniques:

Genetic Algorithm:

The canonical mathematical equation for a Genetic Algorithm is given by:

$$P(t+1) = M(C(S(P(t)))) \quad (1)$$

where:

- $P(t)$ denotes the population at generation t
- $S(\cdot)$ represents the selection operator, which chooses fitter individuals
- $C(\cdot)$ denotes the crossover operator, which recombines selected parents
- $M(\cdot)$ represents the mutation operator, which introduces random variation

Particle Swarm Optimization:

Each particle updates its velocity and position as:

$$v_i^{(t+1)} = \omega v_i^{(t)} + c_1 r_1 (p_i^{\text{best}} - x_i^{(t)}) + c_2 r_2 (g^{\text{best}} - x_i^{(t)}) \quad (2)$$

$$x_i^{(t+1)} = x_i^{(t)} + v_i^{(t+1)} \quad (3)$$

where:

- $x(t)_i$: position (solution vector)
- $v(t)_i$: velocity
- $p_{\text{best}i}$: personal-best position
- g_{best} : global-best position
- ω : inertia weight
- c_1, c_2 : cognitive and social coefficients
- $r_1, r_2 \sim U(0, 1)$: random scalars

Bayesian Optimization:

The Bayesian Optimization process selects the next point by maximizing the acquisition function:

$$x_{t+1} = \arg \max_{x \in \mathcal{X}} a(x; p(f | D_t)) \quad (4)$$

where:

- D_t = observed data up to iteration t
- $p(f | D_t)$ = posterior distribution over the objective function

- $a(x)$ = acquisition function (e.g., Expected Improvement, UCB)

Grid Search CV:

Grid Search CV aims to find the optimal set of hyperparameters θ^* that maximize a chosen evaluation metric across K-fold cross-validation splits:

$$\theta^* = \arg \max_{\theta \in \Theta} \frac{1}{K} \sum_{k=1}^K M(v_{\text{true}}^{(k)}, f(Y_{\text{train}}^{(k)}; \theta)) \quad (5)$$

where:

- Θ = predefined grid of hyperparameter combinations
- θ = one specific hyperparameter combination
- K = number of CV folds
- $f(X^{(k)}; \theta)$ = model trained on k-th training split
- $Y_{\text{train}}^{(k)}$ = training data
- $v_{\text{true}}^{(k)}$ = true labels for the k-th validation split
- $M(\cdot)$ = model evaluation metric (e.g., accuracy)

Random Search:

The probability of successfully finding a good configuration after n random trials is:

$$P(\text{success}) = 1 - (1 - p)^n \quad (6)$$

where:

- p = probability that one random sample is good
- n = number of random samples

These formulations enable algorithms to iteratively search for optimal hyperparameter values using various optimization techniques.

IV. RESULT AND COMPARISON

This section reports the results obtained from various machine learning models and compares their performance across different hyperparameter optimization techniques.

With the base of GA, it identifies top 5 predictive features. Some of them are especially useful for financial practitioners:

- PAY_0 (Repayment Status in Month 1): This is the most important variable. It is indicative of immediate past.
- BILL_AMT1 (Amount of Bill Statement): A large bill relative to credit limits significantly enhances chance for default.
- LIMIT_BAL (Amount of Given Credit): Indicative of the bank's internal risk assessment and the total credit limit to users.
- PAY_AMT1 (Amount of Previous Payment): Sparse payments or some that are small compared to the bill amount are huge red flags.
- AGE: In this particular dataset, the younger demographic segments were statistically more likely to default.

While Genetic Algorithms (GA) and Particle Swarm Optimization (PSO) yield superior accuracy, they demand significantly higher computational resources than traditional search methods. Grid Search suffers from a very high relative execution time due to its slow, exhaustive nature, yet still carries a

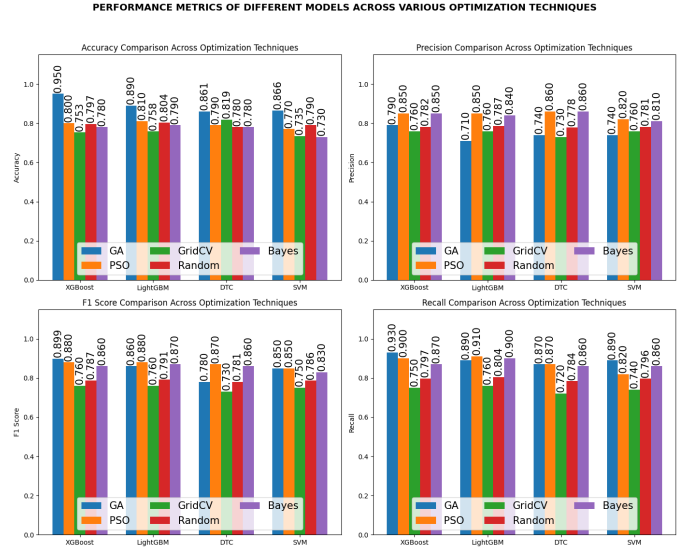


Fig. 5. Performance metrics of models across various optimization techniques

high risk of getting stuck in local optima. On the other hand, Random Search is low in execution time and provides fast, stochastic convergence, but maintains a comparably high risk of local optima. Bayesian Optimization, with moderate speed of execution and medium probabilistic convergence, has low risk of local optima. The higher runtime for GA and PSO is justified by their ability to “evolve” out of local optima and discover more robust hyperparameter configurations that shallower algorithms cannot reach. This computational hardening translates to a better predictive performance and the increased runtime seems justified given that a high accuracy of 95.0% and recall of 93.0% is obtained, which efficient yet shallower algorithms were not able to reach.

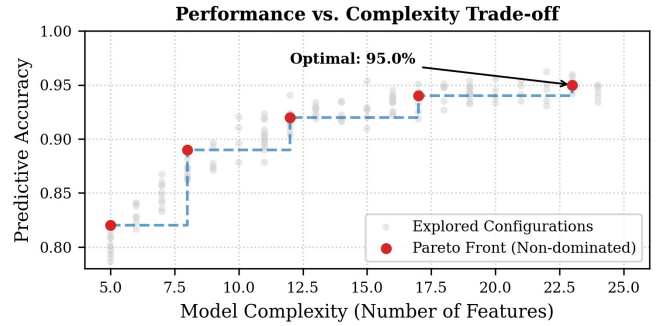


Fig. 6. Pareto Front: Accuracy vs Complexity

This Pareto plot illustrates the contrast between model complexity and predictive accuracy. The optimal solution achieves around 95% accuracy with a set of moderate number of features. Looking at the red points, which represent non dominated solutions, they highlight optimal configurations that are capable of achieving high accuracy with minimal feature complexity. Furthermore, increasing the number of features

would only yield marginal performance gains.

V. FUTURE SCOPE

While this study establishes a foundation for default prediction, real world credit risk is dynamic. To move toward real time risk monitoring, future research should look beyond static snapshots and incorporate ongoing financial behaviors, such as evolving repayment habits and spending trajectories over time. This shift can be effectively captured using temporally-aware deep learning architectures, like attention-based Long Short-Term Memory (LSTM) networks or Temporal Fusion Transformers [26]. Furthermore, as automated lending systems grow more complex, they must remain transparent and fair. To meet regulatory standards and build consumer trust in the banking sector, future iterations must prioritize Explainable AI (XAI). Researchers can explore methods that offer insights as Counterfactual Explanations. These not only clarify the reasoning behind a loan denial but also provide with understandable steps to improve credit standing [27]. Finally, practical deployment in the banking sector requires computational efficiency. Future work could focus on reducing the heavy processing costs associated with feature selection and model tuning by blending the broad exploration of global evolutionary algorithms with the precision of local search methods.

VI. CONCLUSION

This study effectively demonstrated when bio-inspired approaches outperform traditional hyperparameter optimization techniques in solving credit risk assessment problems. According to the extensive evaluation conducted on four machine learning models and five different optimization approaches, the XGBoost model optimized using Genetic Algorithm emerged as the most robust technique for loan default detection, achieving a maximum accuracy of 0.95 and recall of 0.93, respectively. The experimental results clearly indicate that evolutionary techniques, particularly Genetic Algorithm and Particle Swarm Optimization, are superior compared to conventional exhaustive methods such as Grid Search and Random Search, which converged at lower performance levels. Overall, this paper demonstrates that leveraging evolutionary learning in combination with gradient boosting ensemble techniques can lead to significantly more reliable predictions for financial institutions in addressing non-performing assets.

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